MTAP fusion-hotspot benchmark report

Study: msk_impact_50k_2026

Abstract

This report analyzes MTAP gene fusions from the msk_impact_50k_2026 study, covering 4 fusion events. 1/3 fusions with known frame status (33.3%) were in-frame. 1/4 fusions (25.0%) retained the Phosphorylase superfamily domain. Domain retention was tested with Fisher's exact test (p=0.25, not statistically significant at alpha=0.05). No literature benchmark is configured for MTAP.

Results summary

Fusion partner frequency

Fusion-partner frequency was tabulated across 4 analyzed fusion events, identifying 2 distinct fusion partner genes. The most frequent partner was CDKN2A, observed in 3 events.

Domain retention

Phosphorylase superfamily domain retention was tested with Fisher's exact test comparing in-frame fusions against all others; 1/1 (100.0%) of in-frame fusions retained the domain, p=0.25 (not statistically significant at alpha=0.05). A breakpoint-position permutation test produced a corroborating empirical p-value of 0.108911.

Domain disruption

Domain-disruption analysis was skipped: MTAP has no disruption_required_domains configured; domain-disruption analysis was skipped.

Junction concentration

Junction concentration could not be tested for this run: 4 event(s) with a known junction exon inside the transcript; at least 5 are needed

Cutpoint detection

Cutpoint detection scanned 4 mapped breakpoints for the protein position that best separates domain-retained from lost/disrupted fusions; the inferred cutpoint was position 151 aa (exon 6; label-permutation p=0.487512). This boundary is descriptive: retained/lost status follows from breakpoint position, so a clean split exists by construction and the p-value is not evidence of selection. This is -104 aa from the nearest configured domain boundary at 255 aa.

Corroborating confidence statistics

Corroborating confidence statistics compared Domain_retention_flags groups "retained" (n=1) and "not_retained" (n=3). For Frame_status, retained: 1/1 (100.0%, 95% CI 20.7%-100.0%); not_retained: 0/2 (0.0%, 95% CI 0.0%-65.8%). Welch's t-test comparing Tumor_variant_count between groups gave means of 16 vs 26.6667, p=unavailable.

Composite evidence score

Composite evidence score reported: domain_disruption sub-score excluded: the supplied result had no applicable p-value (e.g. disruption_required_domains not configured for this gene).

Exon retention

Exon retention produced results for this run (1 row(s) in Exon_retention); see the accompanying table.

Expression association

Expression association reported: No mRNA expression data was available for MTAP in this cohort; expression-association analysis was skipped.

Genomic position recurrence

Genomic position recurrence produced results for this run (4 row(s) in genomic_bin_recurrence, 0 row(s) in exact_position_recurrence, 0 row(s) in protein_position_genomic_spread); see the accompanying table.

Joint partner

Joint partner reported: MTAP has no gene_pair configured; joint-partner analysis was skipped.

Mechanistic interpretation

Mechanistic interpretation produced results for this run; see the accompanying data for details.

Mutation cooccurrence

Mutation cooccurrence reported: MTAP has no mutual_exclusivity_targets configured; co-occurrence/mutual-exclusivity analysis was skipped.

Window detection

Window detection reported: no candidate window of the configured widths (25, 50, 100, 200) met the minimum-event-count guard (min_events_per_window=4) on both sides of the split

Tables

composite_score tables

composite_evidence_ranking

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty_score	Components_applicable	Composite_score	Rank
CDKN2A	3	0.75	0.09629286886244175	None	None	0.2744647709419452	confidence_certainty, domain_retention, recurrence	0.42541491432277684	1
CDKN2B	1	0.25	0.09629286886244175	None	None	0.2744647709419452	confidence_certainty, domain_retention, recurrence	0.1946456835535461	2

cutpoint_detection tables

cutpoint_scan

cutpoint	n_positive_at_or_below	n_positive_above	n_negative_at_or_below	n_negative_above	odds_ratio	p_value	neg_log10_p_value
12	0	1	1	2	0.0	1.0	-0.0
116	0	1	2	1	0.0	1.0	-0.0
151	0	1	3	0	0.0	0.25	0.6020599913279624

domain_retention tables

frame_domain_contingency_table

1	0
0	3

frame_domain_contingency_tables_by_tumor_type

stratum	table
GB	[0, 0], [0, 1]
IHCH	[0, 0], [0, 1]
LUAD	[1, 0], [0, 0]
SKCM	[0, 0], [0, 1]

domain_retention_descriptives

Domain_key	Domain_name	Quantitative_call_count	Fully_retained_count	Truncated_count	Fully_lost_count	Mean_retained_fraction_among_non_retained_calls
auto_key_domain	Phosphorylase superfamily	4	1	3	0	0.6653061224489796

exon_retention tables

Exon_retention

Exon	Retained_event_count	Total_event_count	Retained_fraction
1	1	4	0.25

frequency tables

Partner_gene_counts

Partner_gene	Event_count	Raw_event_count	Sample_count	Patient_count
CDKN2A	3	3	3	3
CDKN2B	1	1	1	1

genomic_position_recurrence tables

genomic_bin_recurrence

chromosome	bin_start	bin_end	n_events	event_ids
9	21814000	21815000	1	EVT-P-0024526-T01-IM6-6
9	21830000	21831000	1	EVT-P-0067843-T01-IM7-2
9	21842000	21843000	1	EVT-P-0019341-T02-IM6-5
9	21860000	21861000	1	EVT-P-0002886-T01-IM3-8

Per-event results (results.tsv)

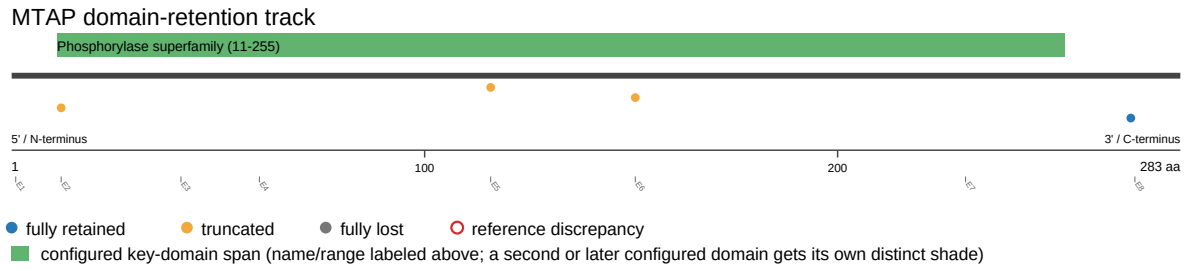
event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	domain_reconstitution	reconstitution_pfam_accession	reconstitution_five_prime_span	reconstitution_three_prime_span	reconstitution_junction_gap_fraction	reconstitution_junction_gap_residues	reconstitution_partner_breakpoint_protein_position	reconstitution_partner_domain_start_aa	reconstitution_partner_domain_end_aa	reconstitution_partner_breakpoint_mid_exon	reconstitution_detail	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-006784 3-T01-IM7-2	P-006784 3-T01-IM7	P-006784 3	Melanoma	SKCM	CDKN2A-MTAP	CDKN2A	out-of-frame	three_prime	21830762	5	116	True	116	True	exact match	disrupted	0.5714285714285714	True	partner_lacks_domain	PF01048					65			False				Phosphorylase superfamily	CDKN2A (NM_058195) - MTAP (NM_002451) rearrangement: c.194-2398:CDKN2A_c.348-7145:MTAPinv|Protein Fusion: out of frame {CDKN2A:MTAP}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	domain_reconstitution	reconstitution_pfam_accession	reconstitution_five_prime_span	reconstitution_three_prime_span	reconstitution_junction_gap_fraction	reconstitution_junction_gap_residues	reconstitution_partner_breakpoint_protein_position	reconstitution_partner_domain_start_aa	reconstitution_partner_domain_end_aa	reconstitution_partner_breakpoint_mid_exon	reconstitution_detail	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-001934 1-T02-IM6-5	P-001934 1-T02-IM6	P-001934 1	Glioma	GB	MTAP-CDKN2A	CDKN2A	out-of-frame	three_prime	21842696	6	151	True	151	True	exact match	disrupted	0.42857142855142855	True	partner_lacks_domain	PF01048					65			False				Phosphorylase superfamily	CDKN2A (NM_000077) - MTAP (NM_002451) rearrangement: c.150+245:CDKN2A_c.450+4687:MTAPinv | Protein Fusion: out of frame {CDKN2A:MTAP}	NA
EVT-P-002452 6-T01-IM6-6	P-002452 6	P-002452 6	Hepatobiliary Cancer	IHCH	MTAP-CDKN2A	CDKN2A	unknown	three_prime	21814419	2	12	True	12	True	exact match	disrupted	0.9959183673469387	True	partner_lacks_domain	PF01048					133			False				Phosphorylase superfamily	CDKN2A (NM_000077) rearrangement: c.457+112:CDKN2A_chr9:g.21814419del | Antisense Fusion	NA

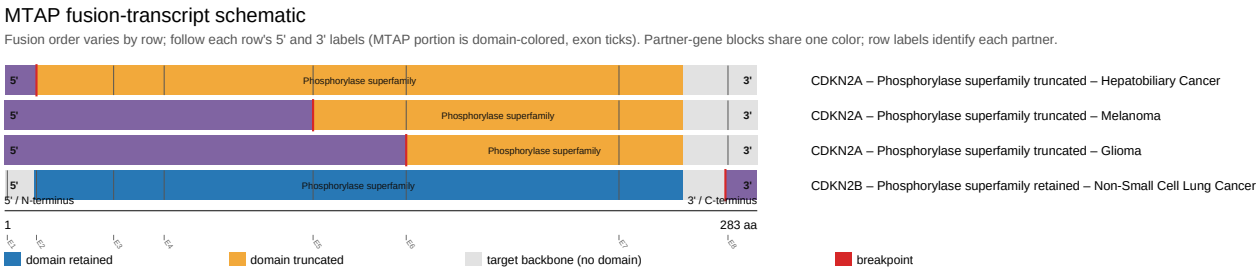
event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	domain_reconstitution	reconstitution_pfam_accession	reconstitution_five_prime_span	reconstitution_three_prime_span	reconstitution_junction_gap_fraction	reconstitution_junction_gap_residues	reconstitution_partner_breakpoint_protein_position	reconstitution_partner_domain_start_aa	reconstitution_partner_domain_end_aa	reconstitution_partner_breakpoint_mid_exon	reconstitution_detail	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-000288 6-T01-IM3-8	P-000288 6	P-000288 6	Non-Small Cell Lung Cancer	LUAD	MTAP-CDKN2B	CDKN2B	in-frame	five_prime	21860421	7	271	True	271	True	exact match	retained	1.0	False													Phosphorylase superfamily			NA | Protein fusion: in frame (MTAP-CDKN2B)	NA

Figures

domain retention outliers



fusion schematic



Partner-block colors are decorative; row labels identify partners.